

Definability in the Infix Order on Words

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Abstract. We develop a theory of (first-order) definability in the infix partial order on words in parallel with a similar theory for the h -quasiorder of finite k -labeled forests. In particular, any element is definable (provided that words of length 1 or 2 are taken as parameters), the first-order theory of the structure is atomic and computably isomorphic to the first-order arithmetic. We also characterize the automorphism group of the structure and show that any arithmetical predicate invariant under the automorphisms of the structure is definable in the structure.

Keywords: Infix order, definability, automorphism, least fixed point, first-order theory, biinterpretability.

1 Introduction

Starting with the classical works of A. Tarski and A.I. Mal'cev, the study of definability in natural structures became a central issue of logic and computation theory. For computation theory, the study of structures on words and trees is the most relevant. In particular, many deep facts on definability and (un)decidability are known for finitely generated free semigroups and groups (see e.g. [Qu46, Ma77, KM06] and references therein).

The study of natural quasiorders on words and trees is also a traditional subject (see e.g. [Th90, Lo97, Ku06] and references therein) with several deep and interesting definability and (un)decidability results.

In [KS07, KS07a, KSZ08, KS09] many results on definability in the quotient structure $(G\mathcal{F}_k; \leq)$ of finite k -labeled forests with the h -quasiorder were established. This line of research is parallel to the popular ongoing study of definability in the degree structures of computability theory, because $(G\mathcal{F}_k; \leq)$ is isomorphic to a natural initial segment of the Wadge degrees of k -partitions of the Baire space [He93, Se07].

* Supported by DFG-RFBR Grant 06-01-04002 and by RFBR grants 08-01-00336, 07-01-00543a.

** Supported by DFG-RFBR Grant 06-01-04002 and by RFBR grant 07-01-00543a.

In this paper we develop a similar complete definability theory for the structure $(A^*; \leq)$ where A^* is the set of words over an arbitrary finite alphabet A with at least two letters and $\leq$ is the infix partial order on A^* defined as follows: $u \leq v$ iff u is an infix of v , i.e. $v = xuy$ for some $x, y \in A^*$. If x (resp. y) in $v = xuy$ may be chosen empty, u is called a prefix (resp. a suffix) of v .

We use some standard notation on words, in particular ε denotes the empty word, $xu = x \cdot u$ denotes the concatenation of words x and u , $|u|$ denotes the length of a word u , $|u|_a$ denotes the number of occurrences of the letter $a \in A$ in the word u , $A^{[m,n]}$ denotes the set $\{u \in A^* \mid m \leq |u| \leq n\}$. We identify letters with the corresponding one-letter words. W.l.o.g. we may assume that $A = A_k = \{0, \dots, k-1\}$ for some $k \geq 2$.

For a given structure $\mathbf{A}$ of signature σ , a predicate on A is *definable* if it is defined by a first-order formula of signature σ (in fact, this definition is not completely precise; to get the well-known precise definition, one has to fix also a suitable list of variables, as in the definition of the operator Γ in the next section). A function on $\mathbf{A}$ is definable if its graph is definable. An element is definable if the corresponding singleton set is definable. A structure is definable if its universe and all signature predicates are definable.

Section 2 recalls some necessary facts from [KS09]. In Section 3 we show that any element of A^* is definable in the $A^{[1,2]}$ -expansion of $(A^*; \leq)$ (i.e. in the expansion obtained by adding to the signature $\{\leq\}$ the constant symbols denoting words of lengths 1 or 2) and characterize the automorphism group of $(A^*; \leq)$. In Section 4 we establish an important technical coding result for the $A^{[1,2]}$ -expansion of $(A^*; \leq)$. In Section 5 we establish our main definability results for the infix order, in particular we show that any arithmetical predicate on A^* which is invariant under the automorphisms of $(A^*; \leq)$ is definable in $(A^*; \leq)$.

The last result implies that the structure $(\omega; +, \cdot)$ is definable in $(A^*; \leq)$ (which improves the result in [Qu46] that $(\omega; +, \cdot)$ is definable in $(A^*; \cdot)$) and that the first-order theory $FO(A^*; \leq)$ of this structure is computably isomorphic to $FO(\omega; +, \cdot)$ (which improves the result in [Ku06] that $FO(A^*; \leq)$ is undecidable). These results show that the infix partial order has the maximal possible complexity with respect to definability and first-order theory which is in contrast with the classical theorem of M. Rabin [Ra69] stating that the prefix (and hence also the suffix) partial order has a decidable first-order (even monadic second-order) theory.

In computability theory, people actively discuss several versions of the so called biinterpretability conjecture stating that some structures of degrees of unsolvability are biinterpretable (in parameters) with $(\omega; +, \cdot)$ (see e.g. [Ni00] and references therein). The conjecture (which seems still open for the most important cases) is considered as in a sense the best possible definability result about degree structures. This paper and the paper [KS09] show that some natural structures on words and forests are biinterpretable (even without parameters) with $(\omega; +, \cdot)$. We believe that our methods could be applied for proving such kind of results for many other structures including those considered in [Ku06].

2 Preliminaries on Gandy Theorem and Definability

Here we recall notions and results from [KS09] which are frequently used below.

Along with first-order definability, we are interested in definability by formulas of special kind related to admissible set theory [Ba75, Er96]. Let σ be a finite signature containing a binary relation symbol $\leq$ and possibly some other relational or constant symbols. *RQ-Formulas* of σ are constructed from the atomic formulas according to the usual rules of first-order logic concerning $\wedge, \vee, \neg, \rightarrow$, the usual rules for the (unbounded) quantification with $\forall, \exists$ and the following additional formation rules for the bounded quantification: if φ is an *RQ*-formula and x, y are variables then the expressions $\forall x \leq y \varphi$ and $\exists x \leq y \varphi$ are *RQ*-formulas. As for the usual first-order formulas, any *RQ*-formula is equivalent to a *special RQ*-formula (i.e., a formula without implications that has negations only on the atomic formulas).

Δ_0 -Formulas of signature σ are constructed inductively according to the following rules: any atomic formula of signature σ is a Δ_0 -formula; if φ and ψ are Δ_0 -formulas then $\neg\varphi$, $(\varphi \wedge \psi)$, $(\varphi \vee \psi)$ and $(\varphi \rightarrow \psi)$ are Δ_0 -formulas; if x, y are variables and φ is a Δ_0 -formula then $\forall x \leq y \varphi$ and $\exists x \leq y \varphi$ are Δ_0 -formulas. Σ -Formulas of signature σ are constructed inductively according to the following rules: any Δ_0 -formula is a Σ -formula; if φ and ψ are Σ -formulas then $(\varphi \wedge \psi)$ and $(\varphi \vee \psi)$ are Σ -formulas; if x, y are variables and φ is a Σ -formula then $\forall x \leq y \varphi$, $\exists x \leq y \varphi$ and $\exists x \varphi$ are Σ -formulas.

A predicate on a σ -structure A is Δ_0 -definable (resp. Σ -definable) if it is defined by a Δ_0 -formula (resp. by a Σ -formula). A predicate on A is Δ -definable if both the predicate and its negation are Σ -definable.

Let P be an n -ary predicate symbol not in σ , φ an *RQ*-formula of signature $\sigma \cup \{P\}$, and $\bar{x} = x_1, \dots, x_n$ a list of variables that includes all free variables of φ . We say that P occurs positively in φ if φ is in negation normal form and has no subformulas $\neg P(y_1, \dots, y_n)$. For any n -ary predicate Q on a σ -structure $\mathbf{A}$, we denote by $(\mathbf{A}, Q)$ the expansion of $\mathbf{A}$ to the $\sigma \cup \{P\}$ -structure where P is interpreted as Q . Then we can define the operator $\Gamma = \Gamma_{\varphi, \bar{x}}$ on the n -ary predicates on A that sends any Q to the predicate

$$\Gamma_{\varphi, \bar{x}}(Q) = \{(a_1, \dots, a_n) \mid (\mathbf{A}, Q) \models \varphi(a_1, \dots, a_n)\}.$$

A σ -structure $\mathbf{A}$ is *bounded* iff $\leq$ is a transitive directed relation (directed means that for all $x, y \in A$ there is $z \in A$ with $x, y \leq z$) and, for any Δ_0 -formula φ , $\mathbf{A} \models \forall x \leq t \exists y \varphi$ implies $\mathbf{A} \models \exists v \forall x \leq t \exists y \leq v \varphi$.

For a Σ -formula φ and a variable u not occurring in φ , let φ^u be the Δ_0 -formula obtained from φ by substituting any occurrence $\exists x$ of unbounded existential quantifier in φ by the corresponding bounded existential quantifier $\exists x \leq u$. The next fact is essentially from [Er96].

Proposition 1. *Let $\mathbf{A}$ be a bounded σ -structure and P an n -ary predicate symbol that occurs positively in a Σ -formula φ with the free variables among $\bar{x}$.*

(i) *The operator Γ is monotone, i.e., $Q \subseteq R \subseteq A^n$ implies $\Gamma(Q) \subseteq \Gamma(R)$.*

- (ii) The operator Γ sends Σ -predicates to Σ -predicates.
- (iii) The symbol P occurs positively in the Σ -formula $\exists u \varphi^u$ which is equivalent to φ in $\mathbf{A}$ and hence induces the same operator Γ .

By Proposition 1(i) and Tarski fixed-point theorem, the operator Γ has the least fixed point denoted by $LFP(\Gamma)$. In general, the least fixed points defined in this way may be complicated. But for some structures $\mathbf{A}$ it turns out that the least fixed point of any Σ -formula φ as above is a Σ -predicate (in this case we say that $\mathbf{A}$ has the *Gandy property*). Let us recall a sufficient condition for $\mathbf{A}$ to have the Gandy property established in [KS09].

We say that a σ -structure $\mathbf{A}$ *admits a Δ -coding of finite sets* if there is a binary Δ -predicate $E(x, y)$ on $\mathbf{A}$ such that $E(x, y)$ implies $x \leq y$ for all $x, y \in A$ and, for all $n < \omega$ and $x_1, \dots, x_n \in A$, there is $y \in A$ with

$$\mathbf{A} \models \forall x (E(x, y) \leftrightarrow x = x_1 \vee \dots \vee x = x_n).$$

Informally, y is considered as a code of the set $\{x \mid E(x, y)\}$. The definition requires the existence of at least one code for any finite subset of A , but does not require that the code is unique.

Observe that, by the axiom of choice, there is a sequence $\{set^n\}_{n < \omega}$ of functions $set^n : A^n \rightarrow A$ which codes the finite sets in the sense that, for all $n < \omega$ and $x, x_1, \dots, x_n \in A$ we have: $x_1, \dots, x_n \leq set^n(x_1, \dots, x_n)$ and $E(x, set^n(x_1, \dots, x_n)) \leftrightarrow x \in \{x_1, \dots, x_n\}$ (in particular, set^0 is an element of A such that $E(x, set^0)$ is false for all $x \in A$).

We call a σ -structure $\mathbf{A}$ *locally finite* if $\{x \mid x \leq y\}$ is finite for each $y \in A$. The next result is useful for understanding definability in some structures.

Theorem 1 ([KS09]). *Let $\mathbf{A}$ be a bounded locally finite σ -structure that admits a Δ -coding of finite sets and a Δ -definable copy of $(\omega; \leq)$. Then $\mathbf{A}$ has the Gandy property.*

Recall that a structure $\mathbf{A}$ equipped with a numbering α (i.e., a surjection from ω onto A) is *arithmetical*, if the equality predicate and all signature predicates are arithmetical modulo α . Obviously, any definable predicate on an arithmetical structure $(\mathbf{A}; \alpha)$ is arithmetical (w.r.t. α) and invariant under the automorphisms of $\mathbf{A}$; we say that $(\mathbf{A}; \alpha)$ has the *maximal definability property* if the converse is also true, i.e., any arithmetical predicate invariant under the automorphisms of $\mathbf{A}$ is definable.

Let again $\mathbf{A}$ be a countable σ -structure and let α be a numbering of A . We say that *the elements of $(\mathbf{A}; \alpha)$ are uniformly Σ -definable* if there is an arithmetical sequence of unary Σ -formulas $\{\psi_n(v_0)\}$ such that ψ_n defines the element $\alpha(n)$ in $\mathbf{A}$ for each $n < \omega$.

Recall (cf. [Ho93, Ni00]) that a structure $\mathbf{B}$ of a finite relational signature τ is *biinterpretable* with a structure $\mathbf{C}$ of a finite relational signature ρ if $\mathbf{B}$ is definable in $\mathbf{C}$ (in particular, there is a bijection $f : B \rightarrow B_1$ on a definable set $B_1 \subseteq C^m$ for some $m \geq 1$ which induces an isomorphism on the τ -structure $\mathbf{B}_1$ definable in $\mathbf{C}$), $\mathbf{C}$ is definable in $\mathbf{B}$ (in particular, there is a similar bijection $g : C \rightarrow C_1$

on a definable set $C_1 \subseteq B^n$ for some $n \geq 1$), the function $g^m \circ f : B \rightarrow B^{nm}$ is definable in $\mathbf{B}$ and the function $f^n \circ g : C \rightarrow C^{mn}$ is definable in $\mathbf{C}$.

Theorem 2 ([KS09]). *Let $(\mathbf{A}; \alpha)$ be an arithmetical σ -structure with the uniformly Σ -definable elements such that $\mathbf{A}$ is bounded, locally finite and admits a Δ -coding of finite sets and a Δ -definable copy $(N; \preceq)$ of $(\omega; \leq)$. Then $\mathbf{A}$ has the maximal definability property and is biinterpretable with $(\omega; +, \cdot)$.*

3 Defining Elements and Characterizing Automorphisms

The first main result of this section is the following

Theorem 3. *The elements of A^* are uniformly Σ -definable in the $A^{[1,2]}$ -expansion of $(A^*; \leq)$.*

Proof.¹ For $x, y \in A^*$, let $S(x, y)$ mean that $x \in \{ay, ya\}$ for some $a \in A$. Since $S(x, y)$ is equivalent to $y < x \wedge \neg \exists z \leq x (y < z < x)$, S is a Δ_0 -predicate.

We show by induction on $|w|$ that any word $w \in A^*$ is Σ -definable in the $A^{[1,2]}$ -expansion of $(A^*; \leq)$. For $|w| \leq 2$ the assertion is obvious, so assume that $|w| \geq 3$ and consider the representation $w = aw_1b$ where $a, b \in A$ and $w_1 \in A^+$. By induction, the words $u = w_1b$ and $v = aw_1$ are Σ -definable. Since for any $c \in A$ the word aac is Σ -definable (the case $c = a$ is easy, and for $c \neq a$ we have: $x = aac$ iff $S(aa, x)$, $c \leq x$ and $ca \not\leq x$), it suffices to check that $x = w$ is equivalent to

$$S(x, u) \wedge S(x, v) \wedge \exists y (S(y, x) \wedge aac \leq y) \quad (1)$$

where c is the first letter of w_1 . Obviously, (1) is true for $x = w$, so it remains to deduce $x = w$ from (1). Let x satisfy (1). By first and second conjuncts in (1), only the following four cases are possible: $x = a_1u = a_2v$ for some $a_1, a_2 \in A$, $x = ub_1 = vb_2$ for some $b_1, b_2 \in A$, $x = a_1u = vb_1$ for some $a_1, b_1 \in A$, $x = a_1v = ub_1$ for some $a_1, b_1 \in A$.

In the first case we have $a_1w_1b = a_2aw_1$, hence $w_1b = aw_1$. Then $a = b$ (otherwise, $w_1 = aw_2b$ for some $w_2 \in A^*$, hence $aw_2bb = aaw_2b$ and so $w_2b = aw_2$ which yields an infinite sequence $w_1 > w_2 > \dots$, a contradiction) and therefore $x \in a^*$. This shows that first and second conjuncts in (1) determine x uniquely, hence $x = w$.

The second case is similar to the first one. In the third case we have $x = a_1w_1b = aw_1b_1$, hence $a_1 = a$, $b = b_1$ and $x = w$.

In the fourth case we have $x = w_1bb_1 = a_1aw_1$. An easy induction on $|w_1|$ shows that $b = a$, $b_1 = a_1$, $w_1 = (a_1a)^n a_1$ in case $|w_1| = 2n + 1$ and $b = a_1$, $b_1 = a$, $w_1 = (a_1a)^{n+1}$ in case $|w_1| = 2n + 2$. Hence, $x = (a_1a)^{n+1} a_1$ in case $|w_1| = 2n + 1$ and $x = (a_1a)^{n+2}$ in case $|w_1| = 2n + 2$. By third conjunct in (1), $aaa_1 \leq y$ for some y with $S(y, x)$. This implies that $a = a_1$, $x \in a^*$ and hence $x = w$.

¹ We thank an anonymous referee for sketching this proof which is shorter than our original proof.

Writing down (recursively) the explicit Σ -definitions of elements in the constructive proof above, we obtain a computable (hence arithmetical) sequence $\{\psi_n(v_0)\}$ of Σ -formulas witnessing that the elements of A^* are uniformly Σ -definable (w.r.t. to the natural numbering of A^*). $\square$

Next we characterize the automorphism group $\text{Aut}(A^*; \leq)$ of the structure $(A^*; \leq)$. We start with the following immediate corollary of the last theorem.

Corollary 1. *Any automorphism of $(A^*; \leq)$ identical on $A^{[1,2]}$ is the identity automorphism.*

Let $\mathbf{S}_k$ denote the symmetric group on k elements $\{0, \dots, k-1\}$ and let $\mathbf{A} \simeq \mathbf{B}$ denote that structures $\mathbf{A}$ and $\mathbf{B}$ are isomorphic. Since for $k = 1$ we have $(A_k^*; \leq) \simeq (\omega; \leq)$, $\text{Aut}(A_k^*; \leq)$ is the trivial one-element group. For $k \geq 2$, along with the identity automorphism e the group $\text{Aut}(A_k^*; \leq)$ has also some other elements, in particular the reverse automorphism r defined by $r(i_1 \dots i_n) = i_n \dots i_1$ for all $n \geq 0$ and $i_1, \dots, i_n < k$ (note that $r \circ r = e$).

Lemma 1. *Let $k \geq 2$ and let f be an automorphism of $(A_k^*; \leq)$ such that $f(i) = i$ and $f(ij) = ji$ for all $i, j < k$. Then $f = r$.*

Proof. The function $f \circ r$ is an automorphism of $(A_k^*; \leq)$ identical on $A_k^{[1,2]}$. By Corollary 1, $f \circ r = e$, hence $f = f \circ r \circ r = r$. $\square$

Theorem 4. *For any $k \geq 2$, $\text{Aut}(A_k^*; \leq) \simeq \mathbf{S}_k \times \mathbf{S}_2$.*

Proof. The restriction map $f \mapsto f|_{A_k}$ is easily seen to be a group homomorphism from $\text{Aut}(A_k^*; \leq)$ onto $\mathbf{S}_k$. We check that the kernel K of this homomorphism coincides with $\{e, r\}$. Obviously, $e, r \in K$. Conversely, let $f \in K$, then $f(i) = i$ for all $i < k$. Since $\{01, 10\}$ is the set of minimal elements in $(\{u \in A_k^* \mid 0, 1 < u\}; \leq)$, $f(01) \in \{01, 10\}$. We distinguish two cases.

Case 1. $f(01) = 01$. Then of course $f(10) = 10$. We show that in fact $f(ij) = ij$ for all $i, j < k$. For $i = j$ this is obvious, so let $i \neq j$. Assume first that $0 \in \{i, j\}$, say $0 = i$ (the case $0 = j$ is treated similarly). As above, $f(0j) \in \{0j, j0\}$. Toward a contradiction, suppose that $f(0j) = j0$. For the word $w = 10j$ we have $10, 0j \leq w$, hence $10 = f(10) \leq f(w)$ and $j0 = f(0j) \leq f(w)$. But $|f(w)| = 3$ (because any automorphism of $(A_k^*; \leq)$ obviously preserves the length of words), hence $f(w)$ cannot be above both $10, j0$; a contradiction.

The case $1 \in \{i, j\}$ is symmetric, so it remains to consider the case $0, 1 \notin \{i, j\}$. Since $f(0i) = 0i$ and $i \in \{i, j\}$, $f(ij) = ij$ by taking i in place of 1 in the above argument. We have shown that f is identical on $A_k^{[1,2]}$. By Corollary 1, $f = e$.

Case 2. $f(01) = 10$. By the argument of case 1 one can show that in fact $f(ij) = ji$ for all $i, j < k$. By Lemma 1, $f = r$. This completes the proof of equality $K = \{e, r\}$.

Let $g \mapsto \tilde{g}$ be the embedding of $\mathbf{S}_k$ into $\text{Aut}(A_k^*; \leq)$ defined by $\tilde{g}(i_1 \dots i_n) = g(i_1) \dots g(i_n)$ for all $n \geq 0$ and $i_1, \dots, i_n < k$ (note that $\tilde{g}|_{A_k} = g$ for each $g \in \mathbf{S}_k$). Then $\tilde{\mathbf{S}}_k = \{\tilde{g} \mid g \in \mathbf{S}_k\}$ and K are subgroups of $\text{Aut}(A_k^*; \leq)$, $\tilde{\mathbf{S}}_k \cap K = \{e\}$, and r commutes with any element of $\tilde{\mathbf{S}}_k$. Then each element of $\text{Aut}(A_k^*; \leq)$ is

uniquely representable in the form $\tilde{g} \circ h$ where $g \in \mathbf{S}_k$ and $h \in K$. Therefore, $\text{Aut}(A_k^*; \leq) \simeq \tilde{\mathbf{S}}_k \times K$. This completes the proof because $\tilde{\mathbf{S}}_k \simeq \mathbf{S}_k$ and $K \simeq \mathbf{S}_2$. $\square$

Remark. Note that $\text{Aut}(A_k^*; \leq)$ is slightly bigger than $\text{Aut}(A_k^*; \cdot) \simeq \mathbf{S}_k$.

4 Coding the Finite Sets

Here we establish the following important technical fact.

Theorem 5. *For any $k \geq 2$, the $A_k^{[1,2]}$ -expansion of $(A_k^*; \leq)$ admits a Δ -coding of finite sets.*

We begin with defining some auxiliary predicates and functions on A_k^* and establishing necessary lemmas.

First note that for all distinct $i, j < k$ the sets i^* , i^*j , ji^* , i^*j^* , ji^*j , and similar sets with $*$ replaced by $+$, are Δ -definable (we use the notation in the style of regular expressions). E.g., $x \in ji^+j$ iff $x \in \{i, j\}^*$, $ji \leq x$, $ij \leq x$, $jj \not\leq x$ and $iji \not\leq x$; by Theorem 3, ji^+j is a Δ -set.

Define the binary Δ_0 -predicates S, L on A_k^* as follows: $S(x, y)$ iff $y < x$ and $\neg \exists z < x(y < z)$, $L(x, y)$ iff $x \leq y$ and $[x, y] = \{z \mid x \leq z \leq y\}$ is linearly ordered under $\leq$. Note that $x \leq x_1 \leq y$ and $L(x, y)$ imply $L(x_1, y)$ and that $L(\varepsilon, x)$ iff $x \in i^*$ for some $i < k$.

Lemma 2. *Let $x < y$ and $y \in A_k^*$ has at least two distinct letters. Then $L(x, y)$ iff $y = uxv$ for unique $u, v \in A_k^*$, and exactly one of u, v is empty. In particular, if $L(x, y)$ then x is a prefix or a suffix of y .*

Proof. From right to left, the assertion is obvious. Conversely, let $L(x, y)$, in particular $y = uxv$ for some $u, v \in A_k^*$. First we check that one of u, v is necessarily empty. Suppose the contrary, then $ix = xj$ for some $i, j < k$. Then $i = j$ (otherwise, $x = ix_1j$ and $ix_1 = x_1j$, hence we obtain an infinite chain $x > x_1 > \dots$ which is a contradiction). Moreover, by a similar argument $x \in i^+$. Since y has two distinct letters, there is x_1 having two distinct letters with $x < x_1 < y$. By the argument in parenthesis we obtain a contradiction. It remains to show that x cannot be simultaneously a prefix and a suffix of y . Suppose the contrary. If $y = xux$ for some nonempty $u \in A_k^*$, one easily finds incomparable elements in $[x, y]$. If there is no such u , x must contain distinct letters and xi is incomparable with jx (where $i, j < k$ satisfy $y = xiv$ and $y = wjx$ for some $v, w \in A_k^*$). $\square$

Let d_l (resp. d_r) be the unary function on A_k^* that deletes the first (resp. the last) letter of a given nonempty word, and let $d_l(\varepsilon) = d_r(\varepsilon) = \varepsilon$.

Lemma 3. *The functions d_l and d_r are Δ -definable.*

Proof. By symmetry, it suffices to prove the assertion for d_l , i.e. to show that the graph $w = d_l(x)$ is a Σ -predicate. Simplifying notation, we stick to the crucial particular case of binary alphabet $A_k = \{0, 1\}$. It suffices to check that $w = d_l(x)$

is equivalent to $x = w = \varepsilon \vee \phi$ where ϕ is the conjunction of the Δ_0 -predicate $S(x, w)$ and disjunction of the following Σ -predicates:

$$\begin{aligned} w \in 0^* \wedge x \in 0^* \cup 10^*, \quad w \in 1^* \wedge x \in 1^* \cup 01^*, \\ w \in 0^+1 \wedge x \in 0^+1 \cup 10^+1, \quad w \notin 0^* \cup 1^* \cup 0^+1 \wedge \psi \end{aligned}$$

where ψ is the following formula:

$$\begin{aligned} \exists y, y_1, t, z (S(y_1, y) \wedge y_1 \in 0^* \wedge S(t, y) \wedge t \in 0^*1 \wedge \\ y \not\leq w \wedge y_1 \not\leq z \wedge L(y, z) \wedge L(t, z) \wedge L(w, z) \wedge L(x, z)). \end{aligned}$$

In case $w \in 0^* \cup 1^* \cup 0^+1$ the equivalence is obvious, so it remains to assume $w \notin 0^* \cup 1^* \cup 0^+1$ and show that $w = d_l(x)$ is equivalent to ψ . Let n be the least number with $0^n \not\leq w$. From left to right, we distinguish two cases.

Case 1. $x = 0w$. Set $z = 0^n x$, then $z = 0^n 0w = 0^n 00^l 1w_1$ for some $l \geq 0$ and $w_1 \in k^*$. Then the values $y = 0^n 00^l$, $y_1 = y0$ and $t = y1$ make formula ψ true.

Case 2. $x = 1w$. Set $y = 0^n$, $y_1 = y0$ and $t = y1$. If $w \in 1A_k^*$, set $z = yx$. Then $z = 0^n 111^l 0w_1$ for some $l \geq 0$ and $w_1 \in k^*$, hence ψ is true. Let now $w \in 0A_k^*$. Since $w \notin 0^* \cup 1^* \cup 0^+1$, either $w = 00^l 10w_1$ or $w = 00^l 11w_1$. Setting $z = y1x = 0^n 1100^l 10w_1$ in the first alternative and $z = yx = 0^n 100^l 11w_1$ in the second alternative, we see that ψ is true.

Conversely, let ψ be true and let y, y_1, t, z be satisfying values, in particular $y = 0^m$ for some $m \geq 1$, $y_1 = y0$ and $t = y1$. Since $S(x, w)$, it suffices to show that w is a suffix of x . Suppose the contrary, then $x = wi$ for some $i < k$. Since $L(y, z)$ and $L(t, z)$, by Lemma 2 y, t cannot both be suffixes of z , hence y is a prefix of z . Since $L(w, z)$, $L(x, z)$ and neither w nor x can be a prefix of z , by Lemma 2 both w, x are suffixes of z . Therefore, w is a suffix of x . $\square$

For any $i < k$, define the binary predicate P_i on A_k^* by

$$P_i(v, x) \leftrightarrow v \in i^+ \wedge v \leq x \wedge iv \not\leq x \wedge \neg L(v, x) \wedge L(v, d_l(x)) \wedge L(v, d_r(x)).$$

Define also the binary predicate Q on A_k^* by

$$Q(v, x) \leftrightarrow v \in 1^+0^+ \wedge v \leq x \wedge 1v \not\leq x \wedge v0 \not\leq x \wedge \neg L(v, x) \wedge L(v, d_l(x)) \wedge L(v, d_r(x)).$$

From Lemmas 2 and 3 it follows that P_i and Q are Δ -predicates with the following properties:

Lemma 4. (i) Let $i < k$, $v, x \in A_k^*$ and x has at least two distinct letters. Then $P_i(v, x)$ iff $v \in i^+$ and $x = vx_1v$ for some $x_1 \in A_k^* \setminus (iA_k^* \cup A_k^*i)$ with $v \not\leq x_1$.
(ii) $Q(v, x)$ iff $v \in 1^+0^+$ and $x = vx_1v$ for some $x_1 \in A_k^* \setminus (0A_k^* \cup A_k^*1)$ with $v \not\leq x_1$.

For any $x \in A_k^*$ and $s \geq |x| + 4$, let $\tilde{x} = 01x0$ and let $p_s(x)$ be the word in $0^*\tilde{x}0^*$ that has exactly s zeros at the beginning and the end.

Lemma 5. For any $x \in A_k^*$ and $s \geq |x| + 4$, $\tilde{x}$ is the greatest element (under $\leq$) in the set of words $z \in A_k^* \setminus (0^*1^* \cup 1^*0^*)$ that are below any of $1^s p_s(\tilde{x}) 1^s$, $1^s \tilde{x} 1^s$, $1^s 0 \tilde{x} 1^s$, $1^s \tilde{x} 0 1^s$.

Proof. Obviously, $\tilde{x}$ satisfies all the conditions on z . It remains to check that $z \leq \tilde{x}$ for any z as in the formulation. Simplifying notation, we stick to the case $k = 2$ and distinguish four cases depending on the first and the last letters of z . If $z \in 0A_k^*0$ then $z \leq 1^s\tilde{x}1^s$ implies $z \leq \tilde{x}$. Assume $z \in 1A_k^*1$ and fix an infix embedding of z into $1^sp_s(\tilde{x})1^s$. If the first 1 of z is in the prefix 1^s of $1^sp_s(\tilde{x})1^s$ then the last 1 of z has to be to the right of the prefix 0^s of $p_s(\tilde{x})$, hence $|z|_0 \geq s$. This contradicts to $z \leq 1^s\tilde{x}1^s$ because the last word has at most $|x| + 2$ zeros. A similar contradiction is obtained in case when the last 1 of z is in the suffix 1^s of $1^sp_s(\tilde{x})1^s$. Hence, $z \leq p_s(\tilde{x})$ and therefore $z \leq \tilde{x}$.

If $z \in 1A_k^*0$ then $z = 1^l00^m1w0$ for some $l, m \geq 0$ and $w \in A_k^*$. Fix infix embeddings of z in $1^s\tilde{x}1^s$ and $1^s0\tilde{x}1^s$. Suppose that in both embeddings the first 1 of z is in the prefixes 1^s , then it has the same position in both prefixes 1^s , hence the 1 before w in z also has the same position in any of $1^s\tilde{x}1^s, 1^s0\tilde{x}1^s$. But this is impossible because of the distinct number of zeros in $1^s\tilde{x}1^s$ and $1^s0\tilde{x}1^s$ between those 1's. Hence, in at least one of the embeddings the first 1 of z is in $\tilde{x}$ and therefore $z \leq \tilde{x}$. The remaining case $z \in 0A_k^*1$ is symmetric to the previous case. $\square$

Next we explain informally our coding of finite sets. We define $set^0() = \varepsilon$ and, for $n \geq 1$ and $x_1, \dots, x_n \in A_k^*$,

$$y = set^n(x_1, \dots, x_n) = 1^sc(x_1) \cdots c(x_n)0^s \quad (2)$$

where $s = \max\{|x_1|, \dots, |x_n|\} + 4$ and $c(x) = p_s(\tilde{x})1^s\tilde{x}1^s0\tilde{x}1^s\tilde{x}01^s$. The minimal infixes of y of the form $1^s0^sw1^s0^s$ coincide with the words $1^sc(x_i)0^s$ for $1 \leq i \leq n$, and they may be recovered from y by Lemma 4(ii). The minimal infixes of $1^sc(x_i)0^s$ of the form 1^sw1^s coincide with $1^sp_s(\tilde{x}_i)1^s, 1^s\tilde{x}_i1^s, 1^s0\tilde{x}_i1^s, 1^s\tilde{x}_i01^s$, and they may be recovered from $1^sc(x_i)0^s$ by Lemma 4(i). From the words $1^sp_s(\tilde{x}_i)1^s, 1^s\tilde{x}_i1^s, 1^s0\tilde{x}_i1^s, 1^s\tilde{x}_i01^s$ we may recover $\tilde{x}_i$ by Lemma 5. Finally, $x_i = d_l d_l d_r(\tilde{x}_i)$.

To realize this idea, we need one more lemma. Let f_0, f_1, g be the unary functions on A_k^* defined as follows: $f_i(x)$ is the largest word in i^* below x , and $g(x) = f_1(x)f_0(x)$. Note that if y is as in (2) then $f_0(y) = 0^s, f_1(y) = 1^s$ and $g(y) = 1^s0^s$. Since $L(1^m, 1^m0^n)$ and $L(0^n, 1^m0^n)$ we easily obtain

Lemma 6. *The functions f_0, f_1, g are Δ -definable.*

Proof of Theorem 5. We have to find a Δ -predicate $E(x, y)$ with the properties stated in Section 2. Let $\exists^{max}t \leq uR(t, \dots)$ abbreviate

$$\exists t \leq u(R(t, \dots) \wedge \neg \exists t_1 \leq u(R(t_1, \dots) \wedge t < t_1)).$$

Let $E(x, y)$ be the Δ -predicate

$$\begin{aligned} \exists u \leq y(Q(g(y), u) \wedge \exists^{max}t \leq u(t \notin (0^*1^* \cup 1^*0^*) \wedge \\ \forall z \leq u(P_1(f_1(y), z) \rightarrow t \leq z) \wedge x = d_l d_l d_r(t))). \end{aligned}$$

From the remarks above it follows that if y is as in (2) then $E(x, y)$ iff $x \in \{x_1, \dots, x_n\}$. $\square$

5 Main Results

Now we are ready to establish the following main result of this paper.

Theorem 6. *The $A^{[1,2]}$ -expansion of $(A^*; \leq)$ has the Gandy property, the maximal definability property and is biinterpretable with $(\omega; +, \cdot)$.*

Proof. The expansion is clearly bounded and locally finite. By Theorem 3, the elements of A^* are uniformly Σ -definable. The structure $(0^*; \leq)$ is a Δ -definable copy of $(\omega; \leq)$. By Theorem 5, the expansion admits a Δ -coding of finite sets. Thus all conditions of Theorems 1 and 2 are satisfied. Conclusions of these theorems give the desired properties. $\square$

We formulate an immediate consequence of the main theorem:

Corollary 2. *The structure $(\omega; +, \cdot)$ is definable in the $A^{[1,2]}$ -expansion of the structure $(A^*; \leq)$. Therefore, the first-order theory of the $A^{[1,2]}$ -expansion of $(A^*; \leq)$ is computably isomorphic to $FO(\omega; +, \cdot)$.*

We conclude the paper with the complete characterization of the definable predicates on $(A^*; \leq)$.

Theorem 7. *For any $k \geq 2$, the structure $(A_k^*; \leq)$ has the maximal definability property.*

Proof. Let $S(x, y)$ denote the predicate “ $y < x$ and $\neg \exists z < x(y < z)$ ” on A_k^* . Let $\{v_a\}_{a \in A_k^{[1,2]}}$ be distinct variables and let

$$\rho = \rho(v_0, \dots, v_{k-1}, v_{00}, \dots, v_{0(k-1)}, \dots, v_{(k-1)0}, \dots, v_{(k-1)(k-1)})$$

be a formula of signature $\{\leq\}$ equivalent to the conjunction of the following formulas:

- $S(v_i, \varepsilon)$ for all $i < k$,
- $\neg(v_i = v_j)$ for all distinct $i, j < k$,
- $S(v_{ij}, v_i) \wedge S(v_{ij}, v_j)$ for all $i, j < k$,
- $\neg(v_{ij} = v_{ji})$ for all distinct $i, j < k$,
- $\exists x(S(x, v_{ij}) \wedge S(x, v_{jl}))$ for all pairwise distinct $i, j, l < k$.

Repeating the proof of Theorem 4 we see that ρ defines in $(A_k^*; \leq)$ the orbit $Orb(\bar{b})$ of the tuple

$$\bar{b} = (0, \dots, k-1, 00, \dots, 0(k-1), \dots, (k-1)0, \dots, (k-1)(k-1))$$

(recall that $Orb(\bar{b}) = \{f(\bar{b}) \mid f \in Aut(A_k^*; \leq)\}$ where $\bar{b} = (b_0, \dots, b_n)$ and $f(\bar{b}) = (f(b_0), \dots, f(b_n))$).

Let $P(\bar{x})$, $\bar{x} = (x_1, \dots, x_n)$, be an n -ary arithmetical predicate on A_k^* which is invariant under the automorphisms of $(A_k^*; \leq)$. We have to show that P is definable in $(A_k^*; \leq)$. W.l.o.g. we assume that variables $x_1, \dots, x_n$ are distinct from the variables v_a above. By Theorem 6, there is a formula $\phi(\bar{x})$ of signature $\{\leq, a\}_{a \in A_k^{[1,2]}}$ that defines P . Let ϕ_1 be the formula of signature $\{\leq\}$ obtained

from ϕ by substituting the variable v_a in place of the constant symbol a for all $a \in A_k^{[1,2]}$. Finally, let θ be the formula obtained from $\rho \wedge \phi_1$ by existential quantification over the variables v_a for all $a \in A_k^{[1,2]}$. Then θ defines P in $(A_k^*; \leq)$. $\square$

The last result implies definability in $(A_k^*; \leq)$ of many interesting predicates on words, in particular of the subword partial order [Ku06]. We think that it would be very complicated to write down an explicit “natural” first-order definition of the subword partial order in $(A_k^*; \leq)$.

A proof similar to the proof of the last theorem establishes the following strengthening of a result in [Ku06].

Corollary 3. *The theory $FO(A^*; \leq)$ is computably isomorphic to $FO(\omega; +, \cdot)$.*

From the last theorem and finiteness of any orbit $Orb(\bar{b})$ we immediately obtain the following important model-theoretic properties of $(A^*; \leq)$.

Corollary 4. *The structure $(A^*; \leq)$ is atomic and minimal.*

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